DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

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Lecture 22: Itô's Lemma

Why Ordinary Calculus Fails · The Extra Term · A Heuristic Derivation

September 23, 2026

Today's Agenda

1. Where ordinary calculus breaks

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3. Worked examples

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2. Heuristic derivation
3. Worked examples
4. The general form and the multiplication table

Learning Objectives

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- State both the simple and general forms.
- Apply it to standard examples.

The Ordinary Chain Rule – and Its Hidden Step

For smooth $x(t)$, Taylor gives

$$\Delta f = f'(x)\Delta x + \tfrac{1}{2}f''(x)(\Delta x)^2 + \cdots$$

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For Brownian motion that step is **invalid**: Week 7 gave $(\mathrm{d}W)^2 = \mathrm{d}t$. The squared term is a *first-order* quantity.

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Side by Side

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Brownian W_t

$\Delta W \sim \sqrt{\Delta t}$

$(\Delta W)^2 \sim \Delta t$ **NOT negligible**

keep $\Rightarrow$ Itô's Lemma

Heuristic Derivation

Expand $f(W_t)$ to second order:

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Itô's Lemma – Simplest Form

$$\mathrm{d}f(W_t) = f'(W_t)\,\mathrm{d}W_t + \tfrac{1}{2}f''(W_t)\,\mathrm{d}t$$

The second term is the **Itô correction**.

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A rigorous proof builds the Itô integral as an L^2 limit – graduate probability material. The Taylor argument is not a proof, but it is the right *mental model*: it correctly locates the extra term and its coefficient $\frac{1}{2}f''$. Every practitioner reasons this way.

Example: $f(W) = W^2$

$f' = 2W$, $f'' = 2$:

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Key Takeaway

$$\int_0^T W_t\,\mathrm{d}W_t = \frac{W_T^2 - T}{2}$$

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Biased high by exactly $T/2$. The Itô integral is a martingale; the naive one isn't.

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The exponential picks up a systematic **upward drift of** $\frac{1}{2}$ per unit time – purely from the Itô correction, even though W_t has no drift at all.

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Remember this – it is exactly what produces the $-\sigma^2/2$ in Lecture 23.

Itô Processes and the General Form

Itô Process

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Itô's Lemma – General Form

$$\mathrm{d}f(X_t, t) = \left(f_t + af_x + \tfrac{1}{2}b^2 f_{xx}\right)\mathrm{d}t + bf_x\,\mathrm{d}W_t$$

Where Each Piece Comes From

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Substituting $\mathrm{d}X = a\,\mathrm{d}t + b\,\mathrm{d}W$ and squaring:

$$(\mathrm{d}X)^2 = a^2\underbrace{(\mathrm{d}t)^2}_{\to 0} + 2ab\underbrace{\mathrm{d}t\,\mathrm{d}W}_{\to 0} + b^2\underbrace{(\mathrm{d}W)^2}_{=\mathrm{d}t} = b^2\,\mathrm{d}t$$

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Collect $\mathrm{d}t$ and $\mathrm{d}W$ terms – done.

The Multiplication Table

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Applying Itô = Taylor expansion + this table.

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The continuous-time echo of Week 5's risk-neutral condition – and the heart of Week 9.

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- Multiplication table: $(\mathrm{d}t)^2 = \mathrm{d}t\,\mathrm{d}W = 0$, $(\mathrm{d}W)^2 = \mathrm{d}t$.
- $\int_0^T W\,\mathrm{d}W = \frac{1}{2}(W_T^2 - T)$ – verified by simulation.

Check Your Understanding

Discuss (2 minutes)

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1. Apply Itô to $f(W) = W^3$.
2. Apply Itô to $f(W, t) = tW$ (don't forget f_t).
3. Why is $\mathbb{E}\left[\int_0^T W \, \mathrm{d}W\right] = 0$, but not for $\frac{1}{2}W_T^2$?

Looking Ahead

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Geometric Brownian Motion –
Lecture 23

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Geometric Brownian Motion – Lecture 23

Including the notorious $-\sigma^2/2$ term.